

Well-being differences and changes in people with somatic and psychiatric disorders or multimorbidities: modelling COVID-19 as a global natural experiment of stress vulnerability

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ABSTRACT

Background Somatic and psychiatric illnesses and multimorbidities increase stress-vulnerability and decrease well-being.

Objective We aimed to identify high-risk disorder profiles for lower well-being and well-being decline under stress, modelling COVID-19 as a natural experiment.

Methods The global Collaborative Outcomes Study on Health and Functioning during Infection Times survey assessed the World Health Organization-Five Well-being Index (WHO-5) (0–100) retrospectively pre-pandemic and intra-pandemically. Mixed-effects models, controlling for demographic/pandemic-related covariates, examined coprimary outcomes: total pre-pandemic WHO-5 and pre-to-intra-pandemic change (estimating β); pre-pandemic WHO-5 <50, new pre-to-intra-pandemic WHO-5 <50 and WHO-5 decrease ≥ 10 (estimating ORs $\pm$ 95% CIs). Fixed effects were self-reported somatic and psychiatric diagnoses and within-and-across multimorbidities.

Findings Altogether, 121 066 adults (female=64.0%, white=69.0%, age=42.0 $\pm$ 15.9 years, countries=155) with illnesses (any=49.8%/somatic=44.8%/psychiatric=16.3%) and multimorbidities (any=32.8%/somatic=27.3%/psychiatric=7.4%/somatic-psychiatric=10.8%) were analysed. Well-being was 2–3-fold worse with psychiatric disorders (*pre-pandemic*: autism-spectrum disorder (β =−7.00 $\pm$ 2.93/OR=1.27–2.65), schizophrenia/schizoaffective disorder (β =−5.39 $\pm$ 3.42/OR=1.09–2.64); *pre-to-intra-pandemic*: major depressive disorder (β =−2.45 $\pm$ 0.89/OR=1.25–1.55), post-traumatic stress disorder (β =−1.98 $\pm$ 1.42/OR=1.19–1.67)) versus somatic disorders (*pre-pandemic*: diarrhoea (β =−3.29 $\pm$ 2.49/OR=1.11–2.15), chronic skin disease (CSD) (β =−2.53 $\pm$ 1.44/OR=1.11–1.65); *pre-to-intra-pandemic*: CSD (β =−2.31 $\pm$ 1.47/OR=1.23–1.75), injury (β =−2.22 $\pm$ 0.99/OR=1.11–

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Specific somatic and psychiatric illnesses as well as multimorbidity may impair well-being and increase stress vulnerability. However, comparative research on a wide spectrum of somatic and psychiatric illnesses and specific multimorbidity profiles in relation to both overall well-being and its decline under stress is lacking.

WHAT THIS STUDY ADDS

⇒ In this large-scale global study modelling COVID-19 as a natural experiment of stress vulnerability, specific psychiatric and somatic illnesses were associated with particularly diminished pre-pandemic well-being as well as pre-to-intra-pandemic well-being decline (skin disease, chronic pain, depressive and anxiety disorders, post-traumatic stress disorder).
 ⇒ Psychiatric diagnoses were associated with 2–3-fold worse pre-pandemic well-being than somatic diagnoses, and well-being outcomes were supra-additively worse with multimorbidity, being up to three times the sum of individual illness effects.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ These findings identify high-risk diagnostic groups warranting prioritised prevention and early intervention during high-stress times and highlight a need for integrated care models to treat multimorbidity beyond addressing individual disorders.

1.40)). Multimorbidities were associated with non-linear/supra-additive adverse outcomes beyond single disorders (*pre-pandemic*: anorexia nervosa+schizophrenia/schizoaffective disorder ($\beta=-27.58\pm12.71$ /OR=1.41–19.94), inflammatory bowel disease+psychosis ($\beta=-24.89\pm10.59$ /OR=2.68–24.25), stroke+osteoarthritis ($\beta=-13.68\pm7.33$ /OR=1.23–6.48); *pre-to-intra-pandemic*: haemorrhoids+obsessive-compulsive disorder ($\beta=-8.86\pm6.54$ /OR=1.22–5.21), binge-eating disorder/bulimia nervosa+social anxiety disorder ($\beta=-6.87\pm5.49$ /OR=0.78–2.85), back pain+retinopathy ($\beta=-6.55\pm6.54$ /OR=1.04–4.69)). Dose–response relationships were observed between the number of disorders and poorer well-being (*pre-pandemic* per diagnosis: psychiatric ($\beta=-1.22\pm0.23$ /OR=1.11–1.17), somatic ($\beta=-0.21\pm0.13$ /OR=1.02–1.05); *pre-to-intra-pandemic* per diagnosis: psychiatric ($\beta=-0.37\pm0.24$ /OR=1.04–1.10)).

Conclusions Poor well-being and further decline under stress were observed particularly with psychiatric illnesses and complex psychiatric and mixed psychiatric-somatic multimorbidity.

Clinical implications The finding of supra-additive multimorbidity burden highlights a need for targeted care models, beyond individual conditions. Vulnerable groups require support during crises, beyond COVID-19.

BACKGROUND

Somatic and psychiatric illnesses represent major global burdens. Non-communicable diseases rank among leading causes of disability-adjusted life years, contributing substantially to global mortality and disability.¹ Psychiatric disorders are equally burdensome, affecting ≥ 1 in 8 people, most commonly depressive and anxiety disorders.² Beyond single illness, multimorbidity is particularly concerning.³ Multimorbidity refers to the co-occurrence of >1 disorder within an individual and can be grouped into somatic multimorbidity, psychiatric multimorbidity and mixed somatic–psychiatric multimorbidity, all of which are associated with greater functional impairment and higher healthcare costs than single conditions.³ Multimorbidity is highly prevalent,³ with somatic and psychiatric disorders showing bidirectional relationships.^{4–6} Depression and anxiety especially increase the risk of chronic medical conditions, while chronic somatic illness is associated with subsequent psychiatric disorders.^{4,5,7}

Somatic and psychiatric disorders also impair patient-reported outcomes (PROs), including quality of life (QoL) and well-being.^{3,5,7–11} Accordingly, PROs are gaining importance in medical research and practice, reflecting a shift beyond symptom reduction towards patient-centred care.⁹ Well-being and health overall also show bidirectional relationships, where illness reduces well-being, while higher well-being is associated with more favourable illness trajectories.^{3,9} A 2025 systematic review ($k=552$) found the lowest well-being versus the general population for depression, cancer, epilepsy, Parkinson's disease, psoriasis and infertility.⁹ Moreover, systematic reviews suggest a dose–response relationship between the number of conditions and poorer PROs, with each additional condition being associated with poorer outcomes.^{3,12}

Somatic and psychiatric illness and multimorbidity may not only decrease well-being but also dysregulate stress reactivity, increasing vulnerability to well-being decline under adversity.^{13–16} In 229 293 adults across 44 countries, every chronic somatic condition, and especially multimorbidity, was associated with higher perceived stress.¹³ Prior research further suggests greater stress sensitivity in patients with psychiatric disorders.^{14–16} The COVID-19 pandemic provided a natural experiment to examine

how illness and multimorbidity shape stress vulnerability.^{17–19} For example, a systematic review and meta-analysis ($k=43$ studies) found increased anxiety and depression in the general population during COVID-19, while pre-existing mental disorders were a consistent risk factor for worse mental health.¹⁹

OBJECTIVE

Prior work has focused on specific disorders or broad multimorbidity indices. However, to our knowledge, no prior study has examined a wide spectrum of illnesses and multimorbidity profiles in relation to both well-being and its decline under stress. COVID-19 provided a natural experiment, acting as a common exogenous stressor across healthy and patient populations.¹⁷ Using data from the Collaborative Outcomes Study on Health and Functioning during Infection Times (COH-FIT), a global survey across >150 countries during COVID-19, we examined well-being across illnesses and multimorbidities.^{20,21} However, COVID-19 was not the focus of this study. Instead, COVID-19 was modelled as a natural experiment of stress vulnerability by controlling for pandemic-related covariates.¹⁷ We hypothesised poorer well-being outcomes would be associated with (1) overall somatic/psychiatric illness and multimorbidity versus healthy comparisons, (2) psychiatric more than somatic disorders,²² (3) multimorbidity more than single illness and (4) similar disorder profiles across pre-pandemic and pre-to-intra-pandemic well-being outcomes.

METHODS

Design and setting

COH-FIT was an anonymous, cross-sectional online survey at the individual level, with participant ratings within the last 2 weeks of taking the survey and retrospectively for the last 2 weeks prior to the onset of the pandemic in that participant's region, translated into 30 languages and conducted across 155 countries in six continents during the COVID-19 pandemic.^{20–22} Data were obtained during the COVID-19 pandemic, from April 2020 to May 2022. Ethical approval was obtained by institutional committees in the study investigator's countries. The design and methods of the COH-FIT have been described in detail previously,²⁰ and information relevant for the present study is described below.

This study is reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines²³ (online supplemental table S1).

Participants and eligibility

Participants were adults aged ≥ 18 years, without an upper age limit, from countries where the COH-FIT was conducted, who provided online informed consent. Participants with incomplete responses to questions relevant to the present analyses were excluded.

Outcomes and assessments

The central measure of this study was the WHO-5 Well-Being Index, which captures well-being across five items: (1) positive mood, (2) vitality, (3) relaxation, (4) restorative sleep and (5) interest and engagement.¹¹ Participants rated each item 'during the last 2 weeks of their regular life' before the pandemic and 'during the last 2 weeks' intra-pandemically on a visual analogue scale from 0–100.^{20,21} Based on these ratings, the coprimary well-being outcomes of this study were defined as: (1) total pre-pandemic WHO-5 (0–100), (2) pre-pandemic WHO-5 <50 (clinically validated threshold for screening for potential

depression),¹¹ (3) total pre-to-intra-pandemic WHO-5 change (pre-to-intra Δ), with models for this outcome being corrected for pre-pandemic WHO-5, (4) pre-to-intra-pandemic clinically relevant decrease ≥ 10 points corrected for pre-pandemic WHO-5, with models for this outcome also being corrected for pre-pandemic WHO-5¹¹ and (5) pre-to-intra-pandemic crossing of the clinically validated threshold if not already below the threshold pre-pandemically (from WHO-5 ≥ 50 pre-pandemic to WHO-5 < 50 intra-pandemic). Multiple complementary well-being outcomes were examined to identify cross-outcome predictors and to capture both total continuous well-being change and validated thresholds.^{11 21}

Predictors in this study were single and multimorbid illness across self-reported somatic and psychiatric diagnoses ('Have you ever been diagnosed with a medical disease/mental health condition by a healthcare professional/doctor or psychologist? Select all that apply'). Somatic diagnoses were chronic pain conditions including chronic abdominal pain, chronic lower back pain, chronic migraine/headache and chronic neck pain, chronic gastrointestinal disorders (CGID) including chronic constipation, chronic diarrhoea, inflammatory bowel disease (IBD) and urinary incontinence, as well as allergy, asthma, cancer, cataract, chronic skin disease (CSD), type 1 diabetes mellitus, type 2 diabetes mellitus (T2DM), emphysema or chronic obstructive pulmonary disease (COPD), haemorrhoids, hypertension, injury, liver disease, myocardial infarction/angina pectoris/coronary heart disease, obesity, osteoarthritis, osteoporosis, peptic ulcer disease (PUD), renal disease, retinopathy, stroke/cerebrovascular disease, thyroid disease and varicose veins of the lower extremities. Psychiatric diagnoses were anorexia nervosa, attention-deficit/hyperactivity disorder (ADHD), autism-spectrum disorder (ASD), bipolar disorder, bulimia nervosa/binge eating disorder, delusional disorder, gambling disorder, generalised anxiety disorder (GAD), major depressive disorder (MDD), obsessive-compulsive disorder (OCD), panic disorder, personality disorder, post-traumatic stress disorder (PTSD), problematic use of alcohol, problematic use of substances other than alcohol, psychosis, schizophrenia/schizoaffective disorder, and social anxiety disorder.

Multimorbidity was defined as (1) somatic multimorbidity (intraindividual ≥ 2 somatic diagnoses), (2) psychiatric multimorbidity (intraindividual ≥ 2 mental diagnoses) and (3) mixed somatic-psychiatric multimorbidity (intraindividual ≥ 1 somatic and ≥ 1 psychiatric diagnosis).

Statistical analysis

Baseline demographic, illness and treatment characteristics were described across the full sample and compared between participants without any diagnosis versus participants with ≥ 1 somatic or psychiatric diagnosis. Categorical measures were compared using χ^2 tests. Continuous measures were compared using Welch's two-sample t-tests. For the primary analyses, we used linear mixed-effects models for continuous outcomes (pre-pandemic WHO-5; pre-to-intra-pandemic WHO-5 change), and logistic mixed-effects models for binary outcomes (pre-pandemic WHO-5 < 50 ; pre-to-intra-pandemic WHO-5 change ≥ 10 ; WHO-5 ≥ 50 pre-pandemic to WHO-5 < 50 intra-pandemic). Across all outcomes, models included single or multimorbid illness predictors as fixed effects, with healthy subjects as the reference group, and a random intercept for country. Modelling pre-to-intra-pandemic WHO-5 changes, adjusted for pre-pandemic WHO-5 ratings, captured the same between-group differences in change as a diagnosis-by-time interaction, while

additionally accounting for regression to the mean. Hence, for continuous well-being outcomes, negative β -coefficients (reported as $\beta \pm$ half-width of the 95% CI) denote more adverse outcomes, indicating lower pre-pandemic WHO-5 and greater pre-to-intra-pandemic WHO-5 decreases in patients versus healthy comparisons; for categorical well-being outcomes, ORs > 1.0 denote more adverse outcomes, indicating higher odds for meeting clinically relevant pre-pandemic or pre-to-intra-pandemic thresholds in patients versus healthy comparisons.

All analyses were corrected for age, sex, race and socioeconomic status (SES) (visual-analogue scale 0–100), and for pandemic-related covariates to examine effects independent of pandemic-related factors: current or prior COVID-19 infection, feeling of isolation and level of governmental restriction. For logistic mixed-effects models, analyses were restricted to multimorbidity dyads with ≥ 10 outcome events to reduce bias and instability in parameter estimation. For single illness analyses, all effects with ≥ 10 events in the categorical outcome and $p < 0.05$ were reported, while for multimorbidity analyses, the top 10 dyads with the greatest effects were reported. For the three coprimary pre-to-intra-pandemic well-being change outcomes results were reported only if significant changes were observed across ≥ 2 outcomes, further reducing the risk of false-positive results.

FINDINGS

Sample demographic, illness and treatment characteristics

COH-FIT comprised 121 066 adults (female=64.0%, age=42.0 $\pm$ 15.9 years, white=69.0%, SES=55.9 $\pm$ 18.7), of whom 60 733 (50.2%) reported no diagnosis while 60 333 (49.8%) reported ≥ 1 somatic or psychiatric diagnosis (table 1). In baseline comparisons, participants with ≥ 1 diagnosis were older (diagnosis=46.1 years vs no diagnosis=38.0 years, $p < 0.0001$), more female (diagnosis=66.1% vs no diagnosis=61.9%, $p < 0.0001$) and more frequently people of white ethnicity (diagnosis=75.8% vs no diagnosis=62.2%, $p < 0.0001$). Conversely, the non-diagnosis group comprised more males (diagnosis=33.2% vs no diagnosis=37.6%) and more participants of African (diagnosis=2.0% vs no diagnosis=3.6%) or Asian descent (diagnosis=13.5% vs no diagnosis=25.8%). Compared with subjects without diagnoses, those with diagnoses also reported more pandemic-related stressors, including more COVID-19 diagnoses ($p = 0.0233$), and more family members being diagnosed with ($p < 0.0001$) or deceased ($p < 0.0001$) due to COVID-19. In unadjusted baseline analyses, pre-to-intra-pandemic WHO-5 decline was observed (pre-pandemic=71.2 $\pm$ 19.8, intra-pandemic=59.4 $\pm$ 24.6), without overall pre-pandemic difference by diagnosis status ($p = 0.12$) and small intra-pandemic differences (diagnosis=58.9 vs no diagnosis=60.0, $p < 0.0001$) (table 1).

Pre-pandemic associations with well-being

In main analyses adjusted for age, sex, race, SES and pandemic-related covariates, participants with any self-reported diagnoses generally had poorer pre-pandemic well-being versus healthy comparisons (table 2). Illnesses and multimorbidities with the poorest well-being outcomes are shown in figure 1. Having any diagnosis was associated with a lower pre-pandemic WHO-5 ($\beta = -0.94 \pm 0.45$, $p < 0.0001$) and higher odds of WHO-5 < 50 (OR=1.12, 95% CI 1.05 to 1.20, $p = 0.0009$). Having more diagnoses was associated with a stepwise effect for the number of diagnoses and lower total WHO-5 ($\beta = -0.36 \pm 0.11$, $p < 0.0001$) and with higher odds of WHO-5 < 50 (OR=1.05, 95% CI 1.03

Table 1 Demographic, illness and treatment characteristics

	Overall, N=121 066	No diagnosis, N=60 733	Any diagnosis, N=60 333	Comparison	
Categorical, n (%), (χ^2 test)				Phi (95% CI)	P value
Gender					
Male	42 891 (35.4)	22 861 (37.6)	20 030 (33.2)	0.05 (0.04 to 0.05)	<0.0001***
Female	77 487 (64.0)	37 582 (61.9)	39 905 (66.1)		
Non-binary	462 (0.4)	189 (0.3)	273 (0.5)		
Transgender or intersex	226 (0.2)	101 (0.2)	125 (0.2)		
Ethnicity					
White	83 498 (69.0)	37 751 (62.2)	45 747 (75.8)	0.17 (0.17 to 0.18)	<0.0001***
African/African-descent	3375 (2.8)	2159 (3.6)	1216 (2.0)		
Hispanic	3366 (2.8)	1360 (2.2)	2006 (3.3)		
Asian	23 795 (19.7)	15 640 (25.8)	8155 (13.5)		
Mixed	5121 (4.2)	2756 (4.5)	2365 (3.9)		
Other	1403 (1.2)	789 (1.3)	614 (1.0)		
Prefer not to answer	508 (0.4)	278 (0.5)	230 (0.4)		
Migrant	5897 (7.4)	3000 (7.8)	2897 (7.1)	0.01 (<0.01 to 0.02)	0.0005***
Urbanicity (inhabitants)					
Village/rural (<10 000)	21 415 (17.7)	11 428 (18.8)	9987 (16.6)	0.04 (0.03 to 0.04)	<0.0001***
Small city/town (10 000–100 000)	30 037 (24.8)	15 095 (24.9)	14 942 (24.8)		
Medium city/town (100 000–500 000)	27 312 (22.6)	13 810 (22.7)	13 502 (22.4)		
Large city/town (over 500 000)	42 302 (34.9)	20 400 (33.6)	21 902 (36.3)		
Currently employed	74 176 (61.3)	37 957 (62.5)	36 219 (60.0)	0.03 (0.02 to 0.03)	<0.0001***
Stressors due to COVID-19					
COVID-19 diagnosis	5805 (17.2)	2675 (16.7)	3130 (17.6)	0.01 (<0.01 to 0.02)	0.0233*
Job lost	6035 (13.0)	3078 (13.7)	2957 (12.4)	0.02 (0.01 to 0.03)	<0.0001***
Family member diagnosed	28 898 (24.0)	13 057 (21.6)	15 841 (26.4)	0.06 (0.05 to 0.06)	<0.0001***
Family member deceased	6329 (5.2)	2664 (4.4)	3665 (6.1)	0.04 (0.03 to 0.04)	<0.0001***
Continuous, mean (SD), (Welch's two-sample t-test)				MD (95% CI)	P value
Age, years (18–114)	42.04 (15.87)	38.03 (14.71)	46.08 (15.97)	8.06 (7.88 to 8.23)	<0.0001***
Socioeconomic status (0–100)	55.86 (18.65)	55.79 (18.65)	55.94 (18.66)	0.15 (–0.06 to 0.36)	0.1656
WHO-5 pre-pandemic (0–100)	71.16 (19.82)	71.25 (19.80)	71.07 (19.84)	0.17 (–0.05 to 0.40)	0.1247
WHO-5 intra-pandemic (0–100)	59.44 (24.58)	59.95 (24.32)	58.93 (24.83)	1.02 (0.74 to 1.29)	<0.0001***
Physical health pre-pandemic (0–100)	78.59 (20.24)	82.01 (18.99)	75.13 (20.86)	6.88 (6.66 to 7.11)	<0.0001***
Physical health intra-pandemic (0–100)	74.02 (22.40)	78.35 (20.82)	69.66 (23.07)	8.68 (8.44 to 8.93)	<0.0001***
Pain pre-pandemic (0–100)	24.17 (26.60)	21.18 (26.29)	27.18 (26.57)	6.00 (5.70 to 6.30)	<0.0001***
Pain intra-pandemic (0–100)	27.84 (27.89)	23.65 (26.77)	32.06 (28.36)	8.40 (8.09 to 8.72)	<0.0001***

Categorical variables were compared using χ^2 tests; continuous variables were compared using Welch's two-sample t-tests. Significance: $p<0.05$ (*), $p<0.005$ (**), $p<0.0005$ (***). Socioeconomic status, physical health and pain were self-rated on a visual analogue scale (0–100).

MD, mean difference; WHO-5, WHO Five Well-Being Index.

to 1.06, $p<0.0001$). Having a somatic diagnosis was associated with lower well-being (WHO-5: $\beta=-0.76\pm0.47$, $p=0.0013$; WHO-5<50: OR=1.11, 95% CI 1.03 to 1.19, $p=0.0036$), with a stepwise effect for the number of somatic diagnoses and poorer well-being (per somatic diagnosis WHO-5: $\beta=-0.21\pm0.13$, $p=0.0012$; WHO-5<50 OR=1.03, 95% CI 1.02 to 1.05, $p=0.0003$). Among somatic diagnoses, the lowest pre-pandemic well-being versus healthy comparisons was observed for chronic diarrhoea, CSD, obesity, coronary heart disease, IBD, asthma, chronic abdominal pain, T2DM, chronic lower back pain, chronic migraine/headache and allergies. Having a psychiatric diagnosis was associated with lower pre-pandemic well-being (WHO-5: $\beta=-2.46\pm0.62$, $p<0.0001$; WHO-5<50 OR=1.31, 95% CI 1.20 to 1.43, $p<0.0001$), with a stepwise effect for the number of psychiatric diagnoses and poorer well-being (per psychiatric diagnosis WHO-5: $\beta=-1.22\pm0.23$, $p<0.0001$; WHO-5<50 OR=1.14, 95% CI 1.11 to 1.17, $p<0.0001$). Among most psychiatric diagnoses, lower pre-pandemic well-being versus healthy comparisons was observed for ASD, schizophrenia/

schizoaffective disorder, bulimia nervosa/binge eating disorder, personality disorder, psychosis, PTSD, MDD, ADHD, anorexia nervosa, GAD, problematic alcohol use, social anxiety disorder, bipolar disorder, panic disorder and OCD.

Participants with multimorbidity generally had poorer pre-pandemic well-being versus healthy comparisons, with greater effects than single illnesses (table 2). Having any multimorbidity was associated with lower pre-pandemic well-being (WHO-5: $\beta=-1.28\pm0.51$, $p<0.0001$; WHO-5<50 OR=1.19, 95% CI 1.10 to 1.28, $p<0.0001$).

Somatic multimorbidity was associated with lower pre-pandemic well-being (WHO-5: $\beta=-0.91\pm0.55$, $p=0.0012$; WHO-5<50 OR=1.15, 95% CI 1.06 to 1.25, $p=0.0007$). Among somatic multimorbidity dyads, the lowest pre-pandemic well-being versus healthy comparisons was observed for stroke+osteoarthritis, chronic diarrhoea+PUD, T2DM+IBD, asthma+chronic diarrhoea, chronic diarrhoea+urinary incontinence, chronic diarrhoea+liver disease, chronic abdominal

Table 2 Associations of somatic and psychiatric diagnoses with pre-pandemic well-being and depression thresholds (WHO-5)

Predictor	pre-pandemic WHO-5 score		pre-pandemic WHO-5<50		
	Overall	Mean (SD), per group	β (95%CI), P	N (%), per group OR (95% CI), P	
Any diagnosis (yes/no), n (%)	60333 (49.8)	71.1 (19.8)	−0.94 (−1.39 to −0.49), <0.0001***	8257 (13.7)	1.12 (1.05 to 1.20), 0.0009**
Number of diagnoses (total), mean (SD)	1.38 (2.09)	–	−0.36 (−0.47 to −0.26), <0.0001***	–	1.05 (1.03 to 1.06), <0.0001***
Any somatic diagnosis, n (%)	54226 (44.8)	71.2 (19.8)	−0.76 (−1.23 to −0.30), 0.0013**	7354 (13.6)	1.11 (1.03 to 1.19), 0.0036**
Number of somatic diagnoses, mean (SD)	1.07 (1.69)	–	−0.21 (−0.34 to −0.08), 0.0012**	–	1.03 (1.02 to 1.05), 0.0003***
Chronic diarrhoea	785 (0.6)	68.8 (22.2)	−3.29 (−5.78 to −0.81), 0.0094*	143 (18.2)	1.55 (1.11 to 2.15), 0.0094*
Chronic skin disease	2474 (2.0)	70.3 (20.4)	−2.53 (−3.97 to −1.10), 0.0006**	360 (14.6)	1.35 (1.11 to 1.65), 0.0032**
Obesity	5585 (4.6)	70.4 (20.2)	−2.12 (−3.13 to −1.11), <0.0001***	827 (14.8)	1.31 (1.14 to 1.52), 0.0002***
Coronary heart disease	1582 (1.3)	71.6 (20.3)	−1.94 (−3.83 to −0.06), 0.0436*	213 (13.5)	1.30 (0.99 to 1.71), 0.0548
Inflammatory bowel disease	2484 (2.1)	70.7 (20.7)	−1.75 (−3.27 to −0.24), 0.0235*	358 (14.4)	1.26 (1.01 to 1.56), 0.0387*
Asthma	7524 (6.2)	70.4 (20.1)	−1.53 (−2.39 to −0.67), 0.0005**	1097 (14.6)	1.21 (1.06 to 1.36), 0.0032**
Chronic abdominal pain	1428 (1.2)	70.0 (20.8)	−1.43 (−3.24 to 0.39), 0.1233	231 (16.2)	1.39 (1.08 to 1.78), 0.0105*
Type 2 diabetes mellitus	3778 (3.1)	71.3 (20.2)	−1.32 (−2.60 to −0.03), 0.0447*	535 (14.2)	1.18 (0.97 to 1.42), 0.0950
Chronic lower back pain	6681 (5.5)	71.0 (20.1)	−1.14 (−2.10 to −0.19), 0.0191*	955 (14.3)	1.24 (1.08 to 1.42), 0.0028**
Chronic migraine/headache	4983 (4.1)	71.1 (19.6)	−1.01 (−2.05 to 0.03), 0.0559	670 (13.4)	1.19 (1.02 to 1.38), 0.0250*
Allergy	16454 (13.6)	70.9 (19.8)	−0.92 (−1.58 to −0.26), 0.0063*	2277 (13.8)	1.12 (1.01 to 1.23), 0.0293*
Any psychiatric diagnosis, n (%)	19741 (16.3)	69.6 (20.4)	−2.46 (−3.08 to −1.84), <0.0001***	3065 (15.5)	1.31 (1.20 to 1.43), <0.0001***
Number of psychiatric diagnoses, mean (SD)	0.30 (0.88)	–	−1.22 (−1.45 to −0.99), <0.0001***	–	1.14 (1.11 to 1.17), <0.0001***
Autism-spectrum disorder	481 (0.4)	66.0 (22.4)	−7.00 (−9.93 to −4.07), <0.0001***	91 (18.9)	1.83 (1.27 to 2.65), 0.0013**
Schizophrenia/schizoaffective disorder	405 (0.3)	68.6 (21.7)	−5.39 (−8.81 to −1.97), 0.0020**	63 (15.6)	1.69 (1.09 to 2.64), 0.0201*
Bulimia nervosa/Binge eating disorder	849 (0.7)	67.1 (21.9)	−5.38 (−7.66 to −3.09), <0.0001***	175 (20.6)	1.96 (1.47 to 2.60), <0.0001***
Personality disorder	1335 (1.1)	67.7 (21.9)	−5.37 (−7.20 to −3.55), <0.0001***	244 (18.3)	1.87 (1.48 to 2.35), <0.0001***
Psychosis	452 (0.4)	67.8 (21.5)	−5.16 (−8.33 to −1.99), 0.0014**	75 (16.6)	1.82 (1.22 to 2.73), 0.0034**
Post-traumatic stress disorder	2508 (2.1)	68.1 (21.5)	−4.80 (−6.18 to −3.41), <0.0001***	484 (19.3)	1.81 (1.52 to 2.16), <0.0001***
Major depressive disorder	8067 (6.7)	68.6 (20.9)	−4.30 (−5.16 to −3.43), <0.0001***	1366 (16.9)	1.54 (1.37 to 1.73), <0.0001***
Attention-deficit/hyperactivity disorder	1384 (1.1)	68.2 (20.6)	−3.99 (−5.66 to −2.31), <0.0001***	230 (16.6)	1.44 (1.14 to 1.81), 0.0019**
Anorexia nervosa	1035 (0.9)	67.9 (21.9)	−3.89 (−5.86 to −1.92), 0.0001***	197 (19.0)	1.61 (1.24 to 2.09), 0.0003***
Generalised anxiety disorder	6036 (5.0)	69.3 (20.7)	−3.52 (−4.51 to −2.54), <0.0001***	963 (16.0)	1.45 (1.27 to 1.66), <0.0001***
Problematic alcohol use	835 (0.7)	68.1 (21.8)	−3.51 (−5.82 to −1.21), 0.0028**	167 (20.0)	1.59 (1.17 to 2.17), 0.0029**
Social anxiety disorder	3194 (2.6)	68.4 (21.4)	−3.34 (−4.60 to −2.08), <0.0001***	569 (17.8)	1.61 (1.37 to 1.90), <0.0001***
Bipolar disorder	1458 (1.2)	69.0 (20.8)	−3.24 (−5.07 to −1.42), 0.0005**	233 (16.0)	1.33 (1.03 to 1.72), 0.0283*
Panic disorder	3678 (3.0)	69.5 (20.6)	−2.47 (−3.72 to −1.22), 0.0001***	578 (15.7)	1.36 (1.15 to 1.62), 0.0004***
Obsessive-compulsive disorder	1286 (1.1)	69.2 (21.8)	−2.31 (−4.30 to −0.32), 0.0232*	216 (16.8)	1.47 (1.12 to 1.92), 0.0053*
Any multimorbidity, n (%)	39692 (32.8)	70.9 (20.0)	−1.28 (−1.79 to −0.76), <0.0001***	5601 (14.1)	1.19 (1.10 to 1.28), <0.0001***
Any somatic multimorbidity	33026 (27.3)	71.1 (19.9)	−0.91 (−1.46 to −0.36), 0.0012**	4553 (13.8)	1.15 (1.06 to 1.25), 0.0007**
Stroke+osteoarthritis	100 (0.1)	69.7 (20.4)	−13.68 (−21.02 to −6.35), 0.0003***	12 (12.0)	2.82 (1.23 to 6.48), 0.0146*
Chronic diarrhoea+PUD	80 (0.1)	65.8 (21.5)	−11.78 (−19.40 to −4.16), 0.0025**	20 (25.0)	3.77 (1.66 to 8.59), 0.0016**
T2DM+IBD	156 (0.1)	67.1 (21.0)	−10.54 (−15.79 to −5.29), <0.0001***	25 (16.0)	2.44 (1.31 to 4.52), 0.0047**
Asthma+chronic diarrhoea	158 (0.1)	66.4 (25.4)	−9.56 (−14.71 to −4.41), 0.0003***	36 (22.8)	3.50 (1.99 to 6.15), <0.0001***
Chronic diarrhoea+urinary incontinence	85 (0.1)	66.1 (23.2)	−9.50 (−16.24 to −2.76), 0.0057*	23 (27.1)	4.14 (2.01 to 8.52), 0.0001***

Continued

Table 2 Continued

Predictor	Overall	pre-pandemic WHO-5 score		pre-pandemic WHO-5<50	
		Mean (SD), per group	β (95%CI), P	N (%), per group	OR (95%CI), P
<i>Chronic diarrhoea+liver disease</i>	62 (0.1)	68.2 (22.5)	-9.04 (-16.66 to -1.42), 0.0200*	12 (19.4)	2.59 (1.08 to 6.25), 0.0336*
<i>CAP+urinary incontinence</i>	102 (0.1)	68.9 (23.9)	-8.98 (-15.43 to -2.53), 0.0063*	24 (23.5)	3.08 (1.50 to 6.32), 0.0022**
<i>IBD+urinary incontinence</i>	142 (0.1)	71.9 (21.2)	-8.96 (-15.32 to -2.61), 0.0057*	22 (15.5)	2.26 (1.06 to 4.83), 0.0356*
<i>T2DM+PUD</i>	127 (0.1)	68.9 (21.1)	-8.94 (-15.90 to -1.98), 0.0118*	26 (20.5)	3.44 (1.60 to 7.39), 0.0016**
<i>Obesity+chronic skin disease</i>	366 (0.3)	66.6 (21.1)	-8.52 (-12.14 to -4.90), <0.0001***	71 (19.4)	2.07 (1.33 to 3.23), 0.0013**
<i>Any psychiatric multimorbidity</i>	8917 (7.4)	68.6 (21.0)	-3.65 (-4.48 to -2.82), <0.0001***	1533 (17.2)	1.54 (1.38 to 1.72), <0.0001***
<i>AN+SCZ</i>	32 (0.0)	54.4 (25.7)	-27.58 (-40.28 to -14.87), <0.0001***	10 (31.2)	5.31 (1.41 to 19.94), 0.0133*
<i>PD+SCZ</i>	63 (0.1)	59.7 (26.4)	-21.44 (-29.56 to -13.32), <0.0001***	18 (28.6)	4.79 (2.04 to 11.25), 0.0003***
<i>BN/BED+PAU</i>	76 (0.1)	57.8 (23.6)	-20.75 (-28.38 to -13.12), <0.0001***	28 (36.8)	7.57 (3.43 to 16.71), <0.0001***
<i>AN+psychosis</i>	49 (0.0)	52.8 (25.9)	-19.61 (-28.38 to -10.85), <0.0001***	18 (36.7)	4.09 (1.59 to 10.51), 0.0034**
<i>OCD+SCZ</i>	53 (0.0)	60.4 (24.8)	-17.13 (-26.66 to -7.61), 0.0004***	14 (26.4)	4.07 (1.47 to 11.26), 0.0069*
<i>PTSD+SCZ</i>	54 (0.0)	60.9 (23.0)	-16.59 (-24.91 to -8.28), <0.0001***	14 (25.9)	3.27 (1.31 to 8.14), 0.0111*
<i>BN/BED+psychosis</i>	36 (0.0)	57.7 (24.1)	-16.37 (-26.21 to -6.53), 0.0011**	12 (33.3)	4.77 (1.68 to 13.52), 0.0033**
<i>AN+ASD</i>	48 (0.0)	59.1 (24.3)	-15.94 (-24.13 to -7.75), 0.0001***	14 (29.2)	3.47 (1.42 to 8.46), 0.0062*
<i>PD+psychosis</i>	101 (0.1)	61.9 (23.7)	-15.18 (-21.54 to -8.81), <0.0001***	24 (23.8)	3.94 (1.98 to 7.83), <0.0001***
<i>ASD+BN/BED</i>	29 (0.0)	57.5 (24.6)	-14.94 (-25.56 to -4.31), 0.0059*	10 (34.5)	5.11 (1.68 to 15.54), 0.0041**
<i>Any mixed multimorbidity</i>	13029 (10.8)	69.6 (20.6)	-2.49 (-3.21 to -1.78), <0.0001***	2071 (15.9)	1.37 (1.24 to 1.51), <0.0001***
<i>IBD+psychosis</i>	27 (0.0)	54.0 (24.0)	-24.89 (-35.47 to -14.30), <0.0001***	11 (40.7)	8.06 (2.68 to 24.25), 0.0002***
<i>IBD+PSU</i>	27 (0.0)	54.4 (20.7)	-21.01 (-35.42 to -6.61), 0.0043**	12 (44.4)	9.39 (2.08 to 42.29), 0.0035**
<i>Chronic constipation+PAU</i>	40 (0.0)	59.3 (25.0)	-19.80 (-29.04 to -10.57), <0.0001***	12 (30.0)	4.89 (1.85 to 12.90), 0.0014**
<i>Cataract+PD</i>	37 (0.0)	58.0 (24.8)	-19.47 (-30.46 to -8.49), 0.0005**	11 (29.7)	7.07 (2.27 to 22.04), 0.0007**
<i>Osteoporosis+PD</i>	38 (0.0)	57.1 (27.2)	-17.30 (-26.83 to -7.78), 0.0004***	13 (34.2)	5.46 (2.01 to 14.82), 0.0009**
<i>Chronic diarrhoea+psychosis</i>	23 (0.0)	55.1 (24.5)	-16.83 (-27.85 to -5.82), 0.0027**	10 (43.5)	7.17 (2.28 to 22.51), 0.0007**
<i>COPD+PD</i>	40 (0.0)	57.8 (23.3)	-16.24 (-24.75 to -7.72), 0.0002***	12 (30.0)	3.73 (1.48 to 9.40), 0.0053*
<i>Stroke+PTSD</i>	45 (0.0)	59.3 (21.9)	-16.03 (-24.78 to -7.28), 0.0003***	14 (31.1)	4.90 (1.96 to 12.28), 0.0007**
<i>Chronic diarrhoea+PAU</i>	31 (0.0)	56.5 (19.5)	-15.57 (-27.05 to -4.08), 0.0079*	10 (32.3)	4.08 (1.19 to 14.05), 0.0258*
<i>Stroke+GAD</i>	52 (0.0)	65.4 (23.3)	-15.10 (-24.94 to -5.26), 0.0026**	11 (21.2)	4.67 (1.65 to 13.21), 0.0036**

All analyses are controlled for age, sex, race and socioeconomic status, current or prior COVID-19 infection, level of isolation and level of governmental restriction. Negative β -coefficients denote lower pre-pandemic WHO-5 patients vs the healthy control group; ORs>1.0 denote higher odds for meeting the clinically relevant pre-pandemic threshold. Significance: $p<0.05$ (*), $p<0.005$ (**), $p<0.0005$ (***).

AN, anorexia nervosa; ASD, autism spectrum disorder; BN/BED, bulimia nervosa/binge eating disorder; CAP, chronic abdominal pain; COPD, chronic obstructive pulmonary disease; GAD, generalised anxiety disorder; IBD, inflammatory bowel disease; OCD, obsessive-compulsive disorder; PAU, problematic alcohol use; PD, personality disorder; PSU, problematic substance use (non-alcohol); PTSD, post-traumatic stress disorder; PUD, peptic ulcer disease; SCZ, schizophrenia/schizoaffective disorder; T2DM, type 2 diabetes mellitus; WHO-5, WHO Five Well-Being Index.

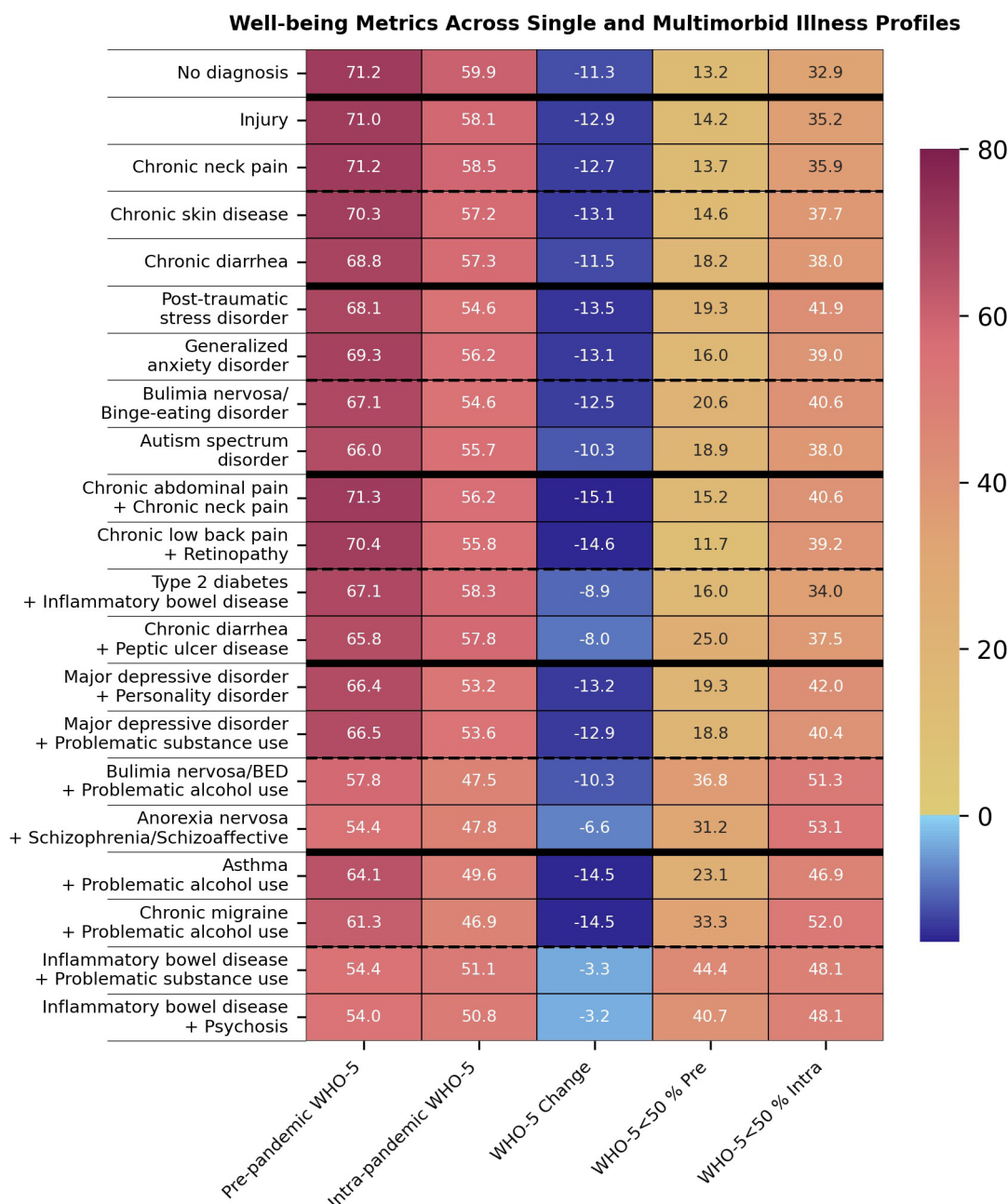


Figure 1 Well-being across single and multimorbid somatic and psychiatric illness profiles before and during the COVID-19 pandemic. Heatmap shows mean World Health Organization-Five Well-Being Index (WHO-5; 0–100) scores pre-pandemic and intra-pandemic, mean pre-to-intra-pandemic WHO-5 change, and the proportion of participants with WHO-5<50 pre-pandemic and intra-pandemic across single diagnoses and multimorbidity profiles with the greatest observed effects. Rows are ordered: somatic single illnesses, psychiatric single illnesses, somatic multimorbidity, psychiatric multimorbidity, mixed somatic-psychiatric multimorbidity. Higher WHO-5 scores indicate better well-being; more negative change values indicate greater decline. WHO-5<50 denotes low well-being. Values represent unadjusted group means or percentages. BED, binge eating disorder.

pain+urinary incontinence, IBD+urinary incontinence, T2DM+PUD, and obesity+CSD.

Psychiatric multimorbidity was also associated with lower pre-pandemic well-being (WHO-5: $\beta = -3.65 \pm 0.83$, $p < 0.0001$; WHO-5 < 50 OR = 1.54, 95% CI 1.38 to 1.72, $p < 0.0001$). Among psychiatric multimorbidity dyads, the lowest pre-pandemic well-being versus healthy comparisons was observed for anorexia nervosa+schizophrenia/schizoaffective disorder, personality disorder+schizophrenia/schizoaffective disorder, bulimia nervosa/binge eating disorder+problematic alcohol use, anorexia

nervosa+psychosis, OCD+schizophrenia/schizoaffective disorder, PTSD+schizophrenia/schizoaffective disorder, bulimia nervosa/binge eating disorder+psychosis, anorexia nervosa+ASD, personality disorder+psychosis and ASD+bulimia nervosa/binge eating disorder.

Having mixed somatic-psychiatric multimorbidity was associated with lower pre-pandemic well-being (WHO-5: $\beta = -2.49 \pm 0.72$, $p < 0.0001$; WHO-5 < 50 OR = 1.37, 95% CI 1.24 to 1.51, $p < 0.0001$). Among mixed multimorbidity dyads, the lowest pre-pandemic well-being versus healthy comparisons was observed for IBD+psychosis,

IBD+problematic use of substances other than alcohol, chronic constipation+problematic alcohol use, cataract+personality disorder, osteoporosis+personality disorder, chronic diarrhoea+psychosis, emphysema/COPD+personality disorder, stroke+PTSD, chronic diarrhoea+problematic alcohol use and stroke+GAD.

Pre-to-intra-pandemic associations with well-being decrease

Participants with self-reported diagnoses generally showed greater pre-to-intra-pandemic well-being decrease versus healthy comparisons (table 3). Having any diagnosis was associated with greater pre-to-intra-pandemic well-being decrease (WHO-5 change: $\beta = -0.54 \pm 0.46$, $p = 0.0226$; WHO-5 decrease ≥ 10 : OR=1.05, 95% CI 1.00 to 1.11, $p = 0.0322$; WHO-5 ≥ 50 pre to WHO-5 <50 intra: OR=1.10, 95% CI 1.04 to 1.17, $p = 0.0007$). Having more diagnoses was associated with a stepwise effect for the number of diagnoses and greater WHO-5 decrease (per diagnosis WHO-5 change: $\beta = -0.13 \pm 0.10$, $p = 0.0119$; WHO-5 ≥ 50 pre to WHO-5 <50 intra: OR=1.02, 95% CI 1.01 to 1.04, $p = 0.0005$). Having a somatic diagnosis was associated with greater pre-to-intra-pandemic well-being decrease (WHO-5 change: $\beta = -0.54 \pm 0.47$, $p = 0.0283$; WHO-5 ≥ 50 pre to WHO-5 <50 intra: OR=1.09, 95% CI 1.03 to 1.16, $p = 0.0036$), with no stepwise effect for each additional somatic diagnosis and worse pre-to-intra-pandemic well-being outcomes. Among somatic conditions, the greatest pre-to-intra-pandemic well-being decreases versus healthy comparisons were observed for CSD, injury, chronic neck pain, chronic lower back pain and allergies.

Having a psychiatric diagnosis was associated with greater pre-to-intra-pandemic well-being decrease (WHO-5 change: $\beta = -1.34 \pm 0.64$, $p < 0.0001$; WHO-5 decrease ≥ 10 : OR=1.14, 95% CI 1.07 to 1.22, $p < 0.0001$; WHO-5 ≥ 50 pre to WHO-5 <50 intra: OR=1.25, 95% CI 1.16 to 1.35, $p < 0.0001$), with a stepwise effect for the number of psychiatric diagnoses and greater well-being decrease (per psychiatric diagnosis WHO-5 change: $\beta = -0.37 \pm 0.24$, $p = 0.0017$; WHO-5 decrease ≥ 10 : OR=1.03, 95% CI 1.00 to 1.05, $p = 0.0333$; WHO-5 ≥ 50 pre to WHO-5 <50 intra: OR=1.07, 95% CI 1.04 to 1.10, $p < 0.0001$). Among psychiatric diagnoses, the greatest pre-to-intra-pandemic well-being decreases versus healthy comparisons were observed for MDD, PTSD, panic disorder and GAD.

Participants with multimorbidity generally showed greater pre-to-intra-pandemic well-being decrease versus healthy comparisons (table 3). Having any multimorbidity was associated with greater pre-to-intra-pandemic well-being decrease (WHO-5 change: $\beta = -0.74 \pm 0.53$, $p = 0.0055$; WHO-5 decrease ≥ 10 : OR=1.09, 95% CI 1.03 to 1.15, $p = 0.0032$; WHO-5 ≥ 50 pre to WHO-5 <50 intra: OR=1.12, 95% CI 1.05 to 1.20, $p = 0.0004$).

Having a somatic multimorbidity was associated with greater pre-to-intra-pandemic well-being decrease (WHO-5 change: $\beta = -0.76 \pm 0.56$, $p = 0.0084$; WHO-5 decrease ≥ 10 : OR=1.07, 95% CI 1.01 to 1.13, $p = 0.0297$; WHO-5 ≥ 50 pre to WHO-5 <50 intra: OR=1.12, 95% CI 1.04 to 1.20, $p = 0.0017$). Among somatic multimorbidity dyads, the greatest pre-to-intra-pandemic well-being decreases versus healthy comparisons were observed for chronic lower back pain+retinopathy, chronic abdominal pain+chronic neck pain, CSD+thyroid disease, allergies+CSD, allergies+injury, allergies+thyroid disease and allergies+asthma. Conversely, several dyads showed associations in the opposite direction for ≥ 1 outcome, including hypertension+osteoporosis, cancer+T2DM and obesity+stroke.

Having a psychiatric multimorbidity was associated with greater pre-to-intra-pandemic well-being decrease (WHO-5 change: $\beta = -1.73 \pm 0.84$, $p < 0.0001$; WHO-5 decrease ≥ 10 : OR=1.15, 95% CI 1.06 to 1.26, $p = 0.0016$; WHO-5 ≥ 50 pre to WHO-5 <50 intra: OR=1.27, 95% CI 1.14 to 1.40, $p < 0.0001$). Among psychiatric multimorbidity dyads, the greatest pre-to-intra-pandemic well-being decreases versus healthy comparisons were observed for bulimia nervosa/binge eating disorder+social anxiety disorder, MDD+problematic substance use, MDD+personality disorder, MDD+problematic alcohol use, MDD+PTSD and GAD+MDD.

Having a mixed somatic-psychiatric multimorbidity was associated with greater pre-to-intra-pandemic well-being decrease (WHO-5 change: $\beta = -1.38 \pm 0.73$, $p = 0.0002$; WHO-5 decrease ≥ 10 : OR=1.13, 95% CI 1.05 to 1.22, $p = 0.0016$; WHO-5 ≥ 50 pre to WHO-5 <50 intra: OR=1.22, 95% CI 1.12 to 1.34, $p < 0.0001$). Among mixed multimorbidity dyads, the greatest pre-to-intra-pandemic well-being decreases versus healthy comparisons were observed for haemorrhoids+OCD, asthma+problematic alcohol use, chronic migraine/headache+problematic alcohol use, haemorrhoids+PTSD, haemorrhoids+MDD, injury+MDD, allergy+PTSD, allergy+MDD, asthma+MDD and hypertension+MDD.

The full list of analysis results is provided in the supplementary materials (online supplemental tables S2–S11).

DISCUSSION

The main findings of this study modelling COVID-19 as a natural experiment of induced stress vulnerability across 121 066 adults in 155 countries are (1) somatic and psychiatric illnesses were consistently associated with poorer pre-pandemic well-being and greater pre-to-intra-pandemic well-being worsening versus healthy comparisons; (2) 2–3-fold poorer well-being outcomes were observed for psychiatric diagnoses than somatic diagnoses; (3) compared with single diagnoses, multimorbidity was associated with marked 2–3-fold poorer well-being outcomes, with psychiatric and mixed multimorbidity showing a dose–response relationship between the number of disorders and well-being; (4) specific diagnoses consistently showed poorer well-being outcomes, particularly conditions characterised by pain (chronic abdominal, low back, neck pain, chronic migraine/headache, injury), social stigma (CSD, CGID, eating disorders), greater risk for comorbidity (heart disease, obesity, T2DM, MDD, anxiety disorders) and poor functioning (CGID, stroke, ASD, psychotic disorders).

These findings can be contextualised by comparison with prior studies of well-being and stress vulnerability. First, single illness was associated with poorer pre-pandemic well-being, consistent with prior evidence.^{8–10 24 25} Lower QoL has been reported for CGID²⁶ and CSD,^{8 9} obesity, heart disease⁶ and chronic pain. Obesity is especially associated with somatic and psychiatric comorbidity,²⁵ and obesity treatment increases QoL while decreasing psychiatric symptoms.²⁵ Among psychiatric illnesses, ASD showed the lowest well-being of all disorders (7-point lower WHO-5 vs healthy comparisons). ASD is associated with low QoL across the life span, primarily being attributed to comorbid depression and impaired social functioning.²⁴ Schizophrenia/schizoaffective disorders were associated with the second-lowest well-being, consistent with their high burden and reduced stress resilience in patients.^{10 14} While present results align with prior research in individual conditions, direct comparative evidence quantifying the relative well-being between multiple somatic and psychiatric illnesses is limited. We found lower well-being for

Table 3 Associations of somatic and psychiatric diagnoses with pre-to-intra-pandemic well-being change

Predictor	WHO-5 score change		WHO-5 decrease (≥10)		pre-pandemic≥50 to intra-pandemic<50	
	Mean (SD), per group	β (95% CI), P	N (%), per group	OR (95% CI), P	N (%), per group	OR (95% CI), P
Any diagnosis, n (%)	-12.1 (21.5)	-0.54 (-1.00 to -0.08), 0.0226*	25101 (42.5)	1.05 (1.00 to 1.11), 0.0322*	14475 (27.8)	1.10 (1.04 to 1.17), 0.0007**
Number of diagnoses, mean (SD)	-	-0.13 (-0.23 to -0.03), 0.0119*	-	1.01 (1.00 to 1.02), 0.2203	-	1.02 (1.01 to 1.04), 0.0005**
Any somatic diagnosis, n (%)	-12.0 (21.4)	-0.54 (-1.01 to -0.06), 0.0283*	22410 (42.2)	1.05 (0.99 to 1.10), 0.0845	12881 (27.5)	1.09 (1.03 to 1.16), 0.0036**
Number of somatic diagnoses, mean (SD)	-	-0.10 (-0.23 to 0.03), 0.1445	-	1.00 (0.99 to 1.02), 0.7115	-	1.02 (1.00 to 1.03), 0.0503
<i>Chronic skin disease</i>						
<i>Injury</i>	-13.1 (22.7)	-2.31 (-3.78 to -0.83), 0.0022**	1058 (44.0)	1.12 (0.96 to 1.31), 0.1438	661 (31.3)	1.47 (1.23 to 1.75), <0.0001***
	-12.8 (21.8)	-2.22 (-3.21 to -1.23), <0.0001***	2832 (43.5)	1.21 (1.10 to 1.35), 0.0002***	1624 (28.4)	1.25 (1.11 to 1.40), 0.0003***
<i>Chronic neck pain</i>	-12.6 (21.8)	-1.53 (-2.81 to -0.26), 0.0185*	1497 (42.9)	1.10 (0.96 to 1.26), 0.1624	906 (29.4)	1.21 (1.03 to 1.41), 0.0197*
<i>Chronic lower back pain</i>	-12.3 (21.3)	-1.30 (-2.28 to -0.32), 0.0092*	2769 (42.2)	1.12 (1.01 to 1.24), 0.0340*	1599 (27.9)	1.10 (0.97 to 1.24), 0.1342
<i>Allergy</i>	-12.6 (22.1)	-0.84 (-1.53 to -0.16), 0.0154*	7009 (43.5)	1.07 (0.99 to 1.15), 0.0688	4110 (29.0)	1.13 (1.04 to 1.23), 0.0041**
Any psychiatric diagnosis, n (%)	-12.8 (22.2)	-1.34 (-1.98 to -0.70), <0.0001***	8537 (44.3)	1.14 (1.07 to 1.22), <0.0001***	5106 (30.6)	1.25 (1.16 to 1.35), <0.0001***
Number of psychiatric diagnoses, mean (SD)	-	-0.37 (-0.61 to -0.14), 0.0017**	-	1.03 (1.00 to 1.05), 0.0333*	-	1.07 (1.04 to 1.10), <0.0001***
<i>Major depressive disorder</i>	-13.0 (22.3)	-2.45 (-3.34 to -1.57), <0.0001***	3519 (44.6)	1.25 (1.14 to 1.37), <0.0001***	2100 (31.3)	1.40 (1.25 to 1.55), <0.0001***
<i>Post-traumatic stress disorder</i>	-13.4 (22.9)	-1.98 (-3.40 to -0.56), 0.0064*	1109 (45.4)	1.20 (1.04 to 1.39), 0.0155*	671 (33.2)	1.41 (1.19 to 1.67), <0.0001***
<i>Panic disorder</i>	-13.3 (22.7)	-1.38 (-2.67 to -0.10), 0.0344*	1590 (44.2)	1.07 (0.93 to 1.22), 0.3460	980 (31.6)	1.21 (1.03 to 1.41), 0.0169*
<i>Generalised anxiety disorder</i>	-13.0 (21.9)	-1.30 (-2.31 to -0.29), 0.0119*	2681 (45.5)	1.18 (1.06 to 1.31), 0.0024**	1593 (31.4)	1.25 (1.11 to 1.42), 0.0003***
Any multimorbidity	-12.3 (21.5)	-0.74 (-1.27 to -0.22), 0.0055*	16703 (43.0)	1.09 (1.03 to 1.15), 0.0032**	9633 (28.3)	1.12 (1.05 to 1.20), 0.0004***
Any somatic multimorbidity	-12.1 (21.4)	-0.76 (-1.32 to -0.19), 0.0084*	13783 (42.6)	1.07 (1.01 to 1.13), 0.0297*	7926 (27.8)	1.12 (1.04 to 1.20), 0.0017**
<i>CLBP+retinopathy</i>	-14.7 (22.0)	-6.55 (-13.09 to -0.01), 0.0496*	51 (44.0)	1.16 (0.58 to 2.33), 0.6763	35 (33.0)	2.21 (1.04 to 4.69), 0.0387*
<i>CAP+TMD</i>	-14.7 (23.7)	-5.66 (-9.26 to -2.06), 0.0021**	157 (48.0)	1.37 (0.94 to 2.00), 0.1026	98 (34.5)	1.54 (1.01 to 2.36), 0.0441*
<i>CSD+TMD</i>	-13.6 (24.5)	-5.06 (-8.39 to -1.74), 0.0028**	168 (42.3)	1.20 (0.85 to 1.70), 0.3059	119 (34.0)	1.97 (1.35 to 2.88), 0.0005***
<i>Allergy+CSD</i>	-13.8 (23.3)	-2.91 (-5.03 to -0.79), 0.0072*	495 (44.7)	1.09 (0.87 to 1.37), 0.4366	332 (33.9)	1.61 (1.26 to 2.06), 0.0001***
<i>Allergy+injury</i>	-13.5 (22.4)	-2.89 (-4.41 to -1.37), 0.0002***	1075 (44.8)	1.31 (1.12 to 1.54), 0.0007**	638 (30.2)	1.39 (1.16 to 1.66), 0.0003***
<i>Allergy+TMD</i>	-12.9 (23.3)	-2.51 (-4.11 to -0.91), 0.0021**	949 (43.8)	1.16 (0.99 to 1.38), 0.0731	585 (30.7)	1.33 (1.10 to 1.60), 0.0028**
<i>Allergy+asthma</i>	-12.3 (22.2)	-1.45 (-2.57 to -0.32), 0.0116*	1822 (43.0)	1.14 (1.02 to 1.28), 0.0268*	1094 (29.5)	1.18 (1.03 to 1.36), 0.0147*
<i>Hypertension+osteoporosis</i>	-11.3 (20.0)	3.63 (0.39 to 6.87), 0.0282*	241 (41.9)	0.86 (0.61 to 1.22), 0.4082	122 (24.2)	0.47 (0.28 to 0.79), 0.0044**
<i>Cancer+T2DM</i>	-8.4 (19.0)	3.73 (0.21 to 7.25), 0.0376*	114 (33.5)	0.54 (0.35 to 0.83), 0.0046**	52 (18.4)	0.57 (0.32 to 1.01), 0.0540
<i>Obesity+stroke</i>	-9.2 (23.3)	6.46 (0.34 to 12.59), 0.0385*	43 (35.8)	0.40 (0.18 to 0.89), 0.0240*	27 (24.8)	0.56 (0.21 to 1.45), 0.2326
Any psychiatric multimorbidity	-13.2 (22.5)	-1.73 (-2.57 to -0.89), <0.0001***	3949 (45.3)	1.15 (1.06 to 1.26), 0.0016**	2343 (31.7)	1.27 (1.14 to 1.40), <0.0001***
<i>BN/BED+SAD</i>	-12.7 (25.1)	-6.87 (-12.36 to -1.39), 0.0141*	77 (45.3)	1.92 (1.08 to 3.42), 0.0274*	46 (33.6)	1.49 (0.78 to 2.85), 0.2329
<i>MDD+PSU</i>	-12.9 (22.3)	-5.91 (-9.90 to -1.92), 0.0037**	112 (46.9)	1.60 (1.05 to 2.42), 0.0271*	63 (31.7)	1.91 (1.20 to 3.05), 0.0064*
<i>MDD+PD</i>	-13.1 (21.8)	-4.32 (-6.85 to -1.78), 0.0008**	294 (41.9)	1.12 (0.86 to 1.46), 0.4112	179 (31.0)	1.37 (1.00 to 1.89), 0.0483*
<i>MDD+PAU</i>	-12.7 (22.4)	-3.44 (-6.80 to -0.08), 0.0445*	185 (45.2)	1.32 (0.93 to 1.87), 0.1236	101 (31.6)	1.88 (1.24 to 2.84), 0.0027**
<i>MDD+PTSD</i>	-12.7 (22.1)	-2.88 (-4.91 to -0.86), 0.0053*	483 (46.0)	1.34 (1.09 to 1.66), 0.0063*	272 (32.6)	1.66 (1.30 to 2.12), <0.0001***
<i>GAD+MDD</i>	-13.2 (22.2)	-2.11 (-3.59 to -0.63), 0.0052*	1091 (45.6)	1.21 (1.04 to 1.41), 0.0167*	660 (32.8)	1.39 (1.16 to 1.66), 0.0004***
Any mixed multimorbidity	-12.8 (22.2)	-1.38 (-2.11 to -0.64), 0.0002***	5597 (43.9)	1.13 (1.05 to 1.22), 0.0016**	3339 (30.5)	1.22 (1.12 to 1.34), <0.0001***
<i>Haemorrhoids+OCD</i>	-13.5 (21.4)	-8.86 (-15.41 to -2.32), 0.0079*	43 (42.6)	1.49 (0.75 to 2.93), 0.2528	31 (36.9)	2.53 (1.22 to 5.21), 0.0121*
<i>Asthma+PAU</i>	-14.8 (21.7)	-8.14 (-13.42 to -2.86), 0.0025**	66 (47.1)	1.92 (1.11 to 3.33), 0.0196*	41 (37.3)	3.02 (1.61 to 5.66), 0.0006**
<i>CM/H+PAU</i>	-13.7 (21.4)	-7.85 (-14.80 to -0.90), 0.0268*	35 (47.9)	1.58 (0.77 to 3.26), 0.2156	17 (34.0)	2.55 (1.00 to 6.50), 0.0498*
<i>Haemorrhoids+PTSD</i>	-14.3 (22.9)	-7.02 (-11.70 to -2.35), 0.0032**	102 (47.4)	1.92 (1.17 to 3.15), 0.0094*	54 (30.2)	1.72 (0.98 to 3.01), 0.0583

Continued

Table 3 Continued

Predictor	WHO-5 score change		WHO-5 decrease (≥ 10)		pre-pandemic ≥ 50 to intra-pandemic < 50	
	Mean (SD), per group	β (95% CI), P	N (%), per group	OR (95% CI), P	N (%), per group	OR (95% CI), P
Haemorrhoids+MDD	-13.6 (23.1)	-5.59 (-8.36 to -2.81), $<0.0001^{***}$	279 (43.5)	1.52 (1.14 to 2.03), 0.0044**	168 (30.9)	1.71 (1.24 to 2.38), 0.0012**
Injury+MDD	-13.1 (23.2)	-5.01 (-7.47 to -2.55), $<0.0001^{***}$	362 (44.5)	1.50 (1.16 to 1.94), 0.0020**	203 (30.4)	1.65 (1.22 to 2.22), 0.0010**
Allergy+PTSD	-13.4 (23.8)	-3.35 (-5.72 to -0.97), 0.0057*	334 (44.6)	1.35 (1.06 to 1.73), 0.0166*	213 (34.7)	1.48 (1.12 to 1.96), 0.0065*
Allergy+MDD	-13.1 (22.7)	-3.28 (-4.89 to -1.67), $<0.0001^{***}$	884 (44.5)	1.26 (1.07 to 1.50), 0.0069*	544 (32.9)	1.67 (1.38 to 2.02), 0.0001***
Asthma+MDD	-12.4 (21.4)	-3.12 (-5.17 to -1.06), 0.0029**	461 (43.2)	1.34 (1.08 to 1.66), 0.0086*	267 (31.0)	1.45 (1.13 to 1.87), 0.0039**
Hypertension+MDD	-12.7 (21.0)	-2.94 (-5.13 to -0.74), 0.0088*	509 (44.5)	1.37 (1.08 to 1.72), 0.0082*	316 (32.6)	1.64 (1.26 to 2.13), 0.0003***

All analyses are controlled for age, sex, race and socioeconomic status, current or prior COVID-19 infection, level of isolation and level of governmental restriction. Negative β -coefficients denote greater pre-to-intra-pandemic WHO-5 decrease patients vs the healthy control group; ORs > 1.0 denote higher odds for meeting clinically relevant pre-to-intra-pandemic thresholds. Significance: $p < 0.05$ (*), $p < 0.005$ (**), $p < 0.0005$ (***).
 ThD, thyroid disease; CAP, chronic abdominal pain; CLBP, chronic low back pain; CM/WH, chronic migraine/headache; CNP, chronic neck pain; CSD, chronic skin disease; GAD, generalised anxiety disorder; MDD, major depressive disorder; OCD, obsessive-compulsive disorder; OR, odds ratio; PAU, problematic alcohol use; PD, personality disorder; PSU, problematic substance use (non-alcohol); PTSD, post-traumatic stress disorder; SAD, social anxiety disorder; T2DM, type 2 diabetes mellitus; WHO-5, WHO Five Well-Being Index.

CGID, CSD, pain and cardiometabolic conditions, but not for liver/renal disease, stroke or cancer. Among psychiatric conditions, no significant well-being decrements were observed for problematic substance use or gambling disorder.

Second, fewer yet similar diagnoses were associated with greater pre-to-intra-pandemic decreases. CSD, beyond having the second-lowest pre-pandemic well-being among somatic disorders, showed the greatest well-being decrease.^{8,9} Pain conditions also showed greater well-being deterioration, potentially from reduced care access during the pandemic.²⁷ Pain may not only be linked to overall psychopathology,²⁷ but eroded stress resilience may moderate the negative impact of pain on psychological outcomes.²⁸

Psychopathology and stress are bidirectionally linked, with mental disorders both resulting from and generating stressful events.^{15,16} Stress was highly prevalent during COVID-19,¹⁹ and previous studies suggest pandemic-related worsening of pre-existing psychiatric conditions.¹⁸ Here, controlling for pandemic-related covariates, MDD, PTSD and anxiety disorders showed greater well-being decrease. MDD and PTSD increased the odds of transitioning from WHO-5 ≥ 50 to < 50 by 40% and 41%, respectively. Meta-analyses indicate heightened stress vulnerability in mood and anxiety disorders,^{15,16} with anxiety potentially mediating later depression through heightened stress reactivity.²⁹ Importantly, differential effects across diagnoses, including negative findings, inform well-being-centred clinical resource allocation.

Third, psychiatric diagnoses were associated with substantially lower pre-pandemic well-being than somatic diagnoses, with 2–3-fold larger effect sizes. Furthermore, most psychiatric diagnoses showed significant decrements in well-being (15/18 psychiatric conditions), contrasting several non-significant results for somatic diagnoses. While direct comparative studies are rare, prior work shows lower well-being and fewer quality-adjusted life years for psychiatric vs somatic illness, particularly depressive and anxiety disorders.⁹ Overall, psychiatric diagnoses were associated with greater pre-to-intra-pandemic well-being decrease than somatic diagnoses. However, pre-to-intra-pandemic well-being outcomes did not differ substantially between top somatic and psychiatric illnesses, indicating stress vulnerability across both illness domains.^{13,18}

Fourth, multimorbidity was consistently associated with poorer well-being versus healthy comparisons, with a dose-response relationship between number of illnesses and poorer well-being outcomes. Consistent with this finding, prior evidence shows poorer PROs across somatic,¹² psychiatric and mixed multimorbidity.^{3,5} Importantly, multimorbidity was associated with disproportionately poorer well-being outcomes than single illnesses. For example, schizophrenia/schizoaffective disorder+anorexia nervosa multimorbidity was associated with ~30-point lower pre-pandemic well-being (scale 0–100), being up to 3× the sum of the respective single-illness effects. Such non-linear, supra-additive effects were most pronounced for complex multimorbidity dyads with large single illness well-being decrements (eg, eating+psychotic disorders, multiple chronic pain conditions, CGID+psychotic disorders). Prior research has identified specific disorder clusters at risk for poorer outcomes.^{5,7} Patients with rare and complex multimorbidities may be more difficult to treat since clinical evidence and care pathways are largely developed for single disorders, neglecting the unique characteristics of uncommon disorder combinations and leaving patients without well-established treatment strategies. Select somatic dyads (T2DM, stroke, obesity) showed less pre-to-intra-pandemic decrease versus healthy comparisons,

likely reflecting floor effects due to already low pre-pandemic well-being, as observed here, and a reduced impact of additional stress in patients exposed to sustained stress from these chronic somatic illnesses.

Fifth, similar disorders showed poorer well-being across single and multimorbidity analyses (chronic pain, CSD, CGID, cardiometabolic; MDD, anxiety, eating, psychotic disorders). However, problematic alcohol and substance use emerged particularly among multimorbidity analyses for psychiatric and mixed multimorbidities, consistent with high comorbidity rates for these disorders.³⁰ Moreover, depression is a central hub of somatic and psychiatric multimorbidity.^{4 6 7 12} However, its impact was most pronounced for pre-to-intra-pandemic well-being decrease rather than for pre-pandemic well-being levels here, with MDD comorbidity in 10/16 significant adverse multimorbidity dyads. This suggests that while depression is a highly prevalent comorbidity in somatic and psychiatric illness, its adverse impact on well-being may be especially pronounced under conditions of heightened stress.^{15 16 18}

The present study has several limitations. First, COH-FIT had a cross-sectional design with retrospective pre-pandemic ratings, potentially introducing recall bias. However, prior COH-FIT analyses indicate acceptable retrospective rating stability.²² Second, diagnoses were self-reported without clinical confirmation. While self-report typically biases towards underestimation, potential misclassification must be acknowledged. Third, binary diagnosis coding limits information on illness severity, duration or treatment status, while increasing interpretability and avoiding overfitting in complex models. Fourth, pandemic exposure, timing and policy responses varied across countries and over time, introducing heterogeneity in stress exposure. However, country-level random effects and adjustment for restriction and infection variables mitigate heterogeneity. Fifth, while prevalence of several diagnoses varies with age, age-stratified subgroup analyses were beyond the scope of this study. Nonetheless, age was adjusted for as a continuous covariate in all models, partly addressing this bias. Sixth, rare multimorbidity combinations were excluded via minimum-event thresholds, potentially excluding high-risk profiles but increasing robustness. Last, COVID-19 was modelled as a natural experiment, rather than examining well-being and its decline under stress through controlled manipulation, limiting causal inference. However, this global real-world setting provides a unique test opportunity that is not otherwise feasible in experimental designs or at this scale.¹⁷ Furthermore, all limitations applied equally to all participants, including those with self-reported diagnoses and healthy comparisons. Despite these limitations, to our knowledge, this is the first study examining a wide spectrum of illnesses and multimorbidity profiles in relation to well-being and stress vulnerability using a global natural experiment.

CLINICAL IMPLICATIONS

In this global natural experiment, the poorest well-being outcomes were observed for psychiatric and mixed multimorbidity. The strongest effects were seen for illness profiles marked by stigma, pain, chronicity, high comorbidity risk and poor functioning. Multimorbidity was prevalent and associated with supra-additive burden beyond individual illnesses. However, most therapies target single disorders rather than addressing the synergistic disease burdens of multimorbidity. Future research is needed to develop treatment strategies tailored specifically to relevant illness combinations. Moreover, PROs should be treated as distinct care targets beyond symptom reduction.⁹ Last,

the present findings inform well-being-centred clinical resource allocation by identifying high-risk illness and multimorbidity profiles, supporting targeted prevention and early intervention for vulnerable patient groups during future large-scale stressors beyond COVID-19.

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Patient consent for publication Consent obtained directly from patients during the initial consent for study participation.

Ethics approval The online survey launches on the COH-FIT website occurred immediately after the first ethics committee/Institutional Review Board (IRB) approval (Aristotle University of Thessaloniki, Greece, 04/27/2020). Afterwards, prior to active local/national investigator outreach and advertisement activities regarding COH-FIT dissemination, approval or waiver (due to the anonymous, observational nature of the study) was obtained from at least one national IRB. All participants consented to participate in the COH-FIT survey. Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer-reviewed.

Data availability statement The dataset for this international survey study includes confidential patient data that cannot be shared in accordance with ethical guidelines and data protection regulations. As part of the consenting process, patients did not consent for their data to be used by other researchers.

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